

The `measuredtex` package*

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Abstract

The `measuredtex` package provides semantic markup for measurement models. It wraps standard L^AT_EX maths environments (`equation`, `align`) so that metadata about each model equation and its constituent terms are written to a sidecar file (`\jobname.models`) during compilation.

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1 Introduction

When writing measurement-model documents it is useful to capture structured metadata alongside the typeset equations. The `measuredtex` package does this transparently: the author writes equations using familiar environments and the package emits a sidecar file.

*This document corresponds to `measuredtex` v0.1, dated 2026-06-02.

2 Usage

Load the package in the usual way:

```
\usepackage{measuredtex}
```

2.1 Model environments

mdeq (*env.*) A wrapper for **equation**. An optional key–value argument accepts the keys **name**, **role**, **source**, and **label**.

```
\begin{mdeq}[name=Ohm's law, label=eq:ohm]
  V = IR
\end{mdeq}
```

mdalign (*env.*) A wrapper for **align**, with the same metadata keys.

2.2 Term definitions

\mddefn **\mddefn**{*label*}{*term*}{*definition*} typesets *term* in inline maths mode and writes a TERM record to the sidecar file. A warning is issued if *label* has not been defined by a prior **\label** command.

2.3 Semantic term macros

\nspace **\nspace**{*text*} renders a namespace identifier in upright roman type.
\assoc **\assoc**{*text*} renders an association identifier in upright roman type.
\tv **\tv**{*text*} renders a term value in italic type.
\qual **\qual**{*text*} or **\qual**{*text*}(*sub*) renders a qualifier, optionally followed by a parenthesised sub-qualifier, both in upright roman type.
\mdterm **\mdterm**[*ns*]{*sym*}[*qual*](*assoc*) composes a decorated term with optional namespace superscript, qualifier subscript, and association argument.

3 Sidecar file format

The file `\jobname.models` contains blocks delimited by header lines:

===== MODEL ===== starts a model record, followed by key–value lines for **name**, **role**, **source**, **label**, and **equation**.

--- TERM --- starts a term record, followed by **eqref**, **term**, and **definition**.

4 Implementation

```
1 \<package>
2 \NeedsTeXFormat{LaTeX2e}
3 \ProvidesExplPackage {measuredtex} {2026-06-02} {0.1}
4   {Semantic markup for measurement models}
5
6 \RequirePackage { amsmath }
```

4.1 Messages

```
7 \msg_new:nnn { measuredtex } { undefined-label }
8 { mddefn:~label~'#1'~is~not~defined. }
```

4.2 Metadata variables

```
9 \tl_new:N \l_md_name_tl
10 \tl_new:N \l_md_role_tl
11 \tl_new:N \l_md_source_tl
12 \tl_new:N \l_md_label_tl
13 \tl_new:N \l_md_model_tl
```

4.3 Output stream

```
14 \iow_new:N \g_md_iow
15
16 \AtBeginDocument
17 {
18   \iow_open:Nn \g_md_iow { \jobname .models }
19 }
20
21 \AtEndDocument
22 {
23   \iow_close:N \g_md_iow
24 }
```

4.4 Metadata keys

```
25 \keys_define:nn { measuredtex / model }
26 {
27   name .tl_set:N = \l_md_name_tl ,
28   role .tl_set:N = \l_md_role_tl ,
29   source .tl_set:N = \l_md_source_tl ,
30   label .tl_set:N = \l_md_label_tl ,
31 }
32
33 \keys_define:nn { measuredtex / mddefn }
34 {
35   model .tl_set:N = \l_md_model_tl ,
36 }
```

4.5 Model-writing logic

`\md_model_write:nn` Clears all metadata keys, applies the supplied key–value pairs, and writes a MODEL block to the sidecar file.

```
37 \cs_new:Npn \md_model_write:nn #1 #2
38 {
39   \tl_clear:N \l_md_name_tl
40   \tl_clear:N \l_md_role_tl
41   \tl_clear:N \l_md_source_tl
42   \tl_clear:N \l_md_label_tl
43
44   \keys_set:nn { measuredtex / model } { #1 }
45
46   \iow_now:Nn \g_md_iow { ===== ~ MODEL ~ ===== }
47 }
```

```

48 \tl_if_empty:NF \l_md_name_tl
49 {
50   \iow_now:Nx \g_md_iow { name: ~ \tl_use:N \l_md_name_tl }
51 }
52
53 \tl_if_empty:NF \l_md_role_tl
54 {
55   \iow_now:Nx \g_md_iow { role: ~ \tl_use:N \l_md_role_tl }
56 }
57
58 \tl_if_empty:NF \l_md_source_tl
59 {
60   \iow_now:Nx \g_md_iow { source: ~ \tl_use:N \l_md_source_tl }
61 }
62
63 \tl_if_empty:NF \l_md_label_tl
64 {
65   \iow_now:Nx \g_md_iow { label: ~ \tl_use:N \l_md_label_tl }
66 }
67
68 \iow_now:Nx \g_md_iow { equation: ~ \tl_to_str:n {#2} }
69 \iow_now:Nn \g_md_iow { }
70 }

```

(End of definition for `\md_model_write:nn`. This function is documented on page ??.)

4.6 Document environments

`mdeq`

```

71 \NewDocumentEnvironment { mdeq } { 0{ } +b }
72 {
73   \md_model_write:nn {#1} {#2}
74   \begin{equation}
75     #2
76     \tl_if_empty:NF \l_md_label_tl
77     {
78       \label { \tl_use:N \l_md_label_tl }
79     }
80   \end{equation}
81 }
82 { }

```

(End of definition for `mdeq`. This function is documented on page ??.)

`mdalign`

```

83 \NewDocumentEnvironment { mdalign } { 0{ } +b }
84 {
85   \md_model_write:nn {#1} {#2}
86   \begin{align}
87     #2
88     \tl_if_empty:NF \l_md_label_tl
89     {
90       \label { \tl_use:N \l_md_label_tl }
91     }

```

```

92   \end{align}
93   }
94   { }

```

(End of definition for `mdalign`. This function is documented on page ??.)

4.7 Term definition command

`\mddefn` #1 is an equation label (existence is checked), #2 is the term content (typeset in maths mode), #3 is the definition text (sidecar only).

```

95 \NewDocumentCommand \mddefn { m m m }
96 {
97   \cs_if_exist:cF { r@#1 }
98   {
99     \msg_warning:nnn { measuredtex } { undefined-label } {#1}
100   }
101   \iow_now:Nn \g_md_iow { ----- ~ TERM ~ ----- }
102   \iow_now:Nx \g_md_iow { eqref: ~ \tl_to_str:n {#1} }
103   \iow_now:Nx \g_md_iow { term: ~ \tl_to_str:n {#2} }
104   \iow_now:Nx \g_md_iow { definition: ~ \tl_to_str:n {#3} }
105   \iow_now:Nn \g_md_iow { }
106   $$#2$
107 }

```

(End of definition for `\mddefn`. This function is documented on page ??.)

4.8 Semantic term macros

These use conventional L^AT_EX 2_ε syntax, so we leave `expl3` catcodes temporarily.

```

108 \ExplSyntaxOff

```

`\nspace`

```

109 \NewDocumentCommand{\nspace}{m}{%
110   \mathrm{#1}%
111 }

```

(End of definition for `\nspace`. This function is documented on page ??.)

`\assoc`

```

112 \NewDocumentCommand{\assoc}{m}{%
113   \mathrm{#1}%
114 }

```

(End of definition for `\assoc`. This function is documented on page ??.)

`\tv`

```

115 \NewDocumentCommand{\tv}{m}{%
116   \mathit{#1}%
117 }

```

(End of definition for `\tv`. This function is documented on page ??.)

`\qual`

```
118 \NewDocumentCommand{\qual}{m d()}{%
119   \mathrm{#1}\IfValueT{#2}{(\mathrm{#2})}%
120 }
```

(End of definition for `\qual`. This function is documented on page ??.)

`\mdterm`

```
121 \NewDocumentCommand{\mdterm}{o m o d()}{%
122   \IfValueT{#1}%
123     {\}\nospac{#1}}}%
124   {#2}%
125   \IfValueT{#3}%
126     {\_{\qual{#3}}}%
127   \IfValueT{#4}%
128     {\(\assoc{#4})}%
129 }
```

(End of definition for `\mdterm`. This function is documented on page ??.)

```
130 \ExplSyntaxOn
```

```
131 \endinput
```

```
132 \</package>
```

Change History

v0.1

General: Initial public version. 1

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The italic numbers denote the pages where the corresponding entry is described, numbers underlined point to the definition, all others indicate the places where it is used.

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